Macroeconomic Impact of the 2026 Gulf Crisis

A Panel Data Analysis of EU3 Countries
(France, Germany, Italy)

Econometrics Report

January 2023 - June 2026 (Forecasted)

Crisis Onset: February 27, 2026

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Panel Data Econometrics | ARIMA Forecasting | Fixed Effects & Robust Inference

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1. Executive Summary

This report investigates the macroeconomic impact of the 2026 Gulf Crisis on the three largest euro-area economies: France, Germany, and Italy (EU3). Using a balanced panel dataset of 126 observations (3 countries x 42 months, January 2023 through June 2026), we estimate the transmission of oil-price shocks and exchange-rate movements to inflation and industrial production. The three post-crisis months (March-May 2026) are observed data; the subsequent three months (April-June 2026) are extended via ARIMA forecasting for the main variables, with policy rates projected via Last Observation Carry-Forward (LOCF) given their step-function nature.

Key Findings

1. Exchange-rate pass-through dominates: Log(NEER) = 0.908 ($p < 2e\text{-}16$) in the FE CPI model, meaning a 1% appreciation of the nominal effective exchange rate reduces log(CPI) by approximately 0.91%.

2. Oil-price pass-through is statistically insignificant: oil_pct_change coefficient = 2.58e-06 ($p = 0.982$) under robust standard errors, contradicting the intuitive expectation of direct oil-to-inflation transmission.

3. Month-seasonality drives growth: FE+Month-Dummies model for IPI achieves R-squared = 0.649, versus 0.071 without month dummies. The August dummy (-0.245, $p = 0.003$) captures the European industrial shutdown.

4. Country-level NEER heterogeneity: Germany (1.14) > France (1.02) > Italy (0.62), indicating Germany is most sensitive to euro appreciation.

5. Post-crisis dummy insignificant for CPI (-0.006, $p = 0.372$) and IPI (0.007, $p = 0.671$), suggesting the crisis has not yet produced a statistically detectable level shift in either variable as of March 2026.

Statement of Equal Contribution

All authors – Yassine Essaadi, Amine Bessaad, Atef Hammami, and Ayoub El Atri – contributed equally to every stage of this project. This includes data collection from official sources (OECD, Eurostat, EIA, BIS, FRED), data cleaning and processing, ARIMA forecasting, panel data econometric analysis (Pooled OLS, Fixed Effects, Random Effects with Wallace-Hussain, Two-Way Fixed Effects, and month-dummy specifications), diagnostic testing, robustness checks (Arellano HC3 and double-clustered standard errors), and the writing of both the final econometrics report and the methodology report. Each author reviewed and approved the final submission.

2. Introduction

2.1 Background and Motivation

The Gulf region remains the world's most critical source of crude oil supply. Any geopolitical disruption in this area has historically triggered sharp increases in oil prices, with cascading effects on inflation, output, and financial markets worldwide. The 2026 Gulf Crisis, erupting on February 27, 2026, represents the most significant such disruption since the 1990-91 Gulf War. Brent crude surged from approximately $75 per barrel in early 2026 to over $100 within weeks, echoing the supply-shock dynamics observed during earlier episodes. For the euro area, the transmission channels are particularly complex: while the direct effect of higher oil prices on import costs is straightforward, the indirect effects through the exchange rate, monetary policy stance, and business confidence create a web of interactions that require careful econometric disentangling.

The three largest euro-area economies, France, Germany, and Italy, share a common monetary policy under the European Central Bank (ECB) and a common currency, yet they differ substantially in their energy mix, industrial structure, and trade exposure. Germany's export-oriented manufacturing sector is highly sensitive to exchange-rate fluctuations, while Italy's reliance on imported energy makes it more vulnerable to oil-price shocks. France occupies an intermediate position, with a more diversified economy and a significant nuclear energy component that partially insulates it from oil-supply disruptions. Understanding how these structural differences mediate the crisis impact is central to the analysis.

2.2 Research Questions

This study addresses three primary research questions. First, to what extent do oil-price changes pass through to consumer prices in the EU3, and does this pass-through differ from the exchange-rate channel? Second, how does the Gulf Crisis affect industrial production, and does seasonality play a confounding role? Third, are there significant cross-country differences in the transmission mechanisms, and do supplementary economies (the United States, Saudi Arabia, Turkey) exhibit qualitatively different patterns?

2.3 Scope and Contributions

The contribution of this report is threefold. Methodologically, we employ a hybrid approach combining observed panel data (39 months) with ARIMA-forecasted extensions (3 months) and LOCF for policy rates, enabling the inclusion of post-crisis observations while maintaining forecast integrity. Empirically, we provide a comprehensive battery of panel estimators (POLS, FE, RE, TWFE, FE+Month-Dummies) with robust standard errors (Arellano HC3 and double-clustered), ensuring the results are not driven by heteroskedasticity or serial correlation. Comparatively, we supplement the EU3 panel with individual OLS regressions for the US, Saudi Arabia, and Turkey, offering a cross-group perspective on the crisis transmission.

3. Literature Review

The macroeconomic effects of oil-price shocks have been studied extensively since Hamilton's (1983) seminal work establishing the asymmetric relationship between oil prices and GDP growth. Hooker (2002) demonstrated that the oil-price pass-through to inflation has declined structurally since the 1980s, a finding attributed to improved monetary policy credibility and the anchoring of inflation expectations. For the euro area specifically, Kilian (2008) distinguished between demand-driven and supply-driven oil shocks, showing that only the latter produce stagflationary outcomes. More recently, Peersman and Van Robays (2012) found that exchange-rate movements can amplify or dampen oil-shock transmission depending on the invoicing currency of trade contracts.

The exchange-rate pass-through (ERPT) literature, pioneered by Goldberg and Knetter (1997) and advanced by Campa and Goldberg (2005), establishes that ERPT to import prices is incomplete but significant, averaging around 0.6 within one year for advanced economies. For consumer prices, the pass-through is lower, typically 0.1-0.3, due to distribution margins and local-cost components. However, these estimates vary considerably across countries and time periods. The European context is particularly interesting because the ECB's inflation-targeting regime may have further reduced ERPT, while the euro's status as a global reserve currency creates additional portfolio-rebalance channels.

Panel data methods for macroeconomic analysis have evolved from simple pooled regressions to sophisticated estimators addressing cross-sectional dependence and heterogeneous dynamics. Pesaran (2006) proposed the Common Correlated Effects (CCE) estimator for panels with unobserved common factors, while Baltagi (2021) provides a comprehensive treatment of the modern panel econometrics toolkit. The present study follows the established sequence of POLS-FE-RE estimation, augmented by Hausman testing and robust inference, which remains the standard approach for moderate-N, moderate-T panels in applied macroeconomics.

4. Data Description

4.1 Data Sources and Construction

The dataset comprises monthly observations for six countries from January 2023 to June 2026. The main panel consists of France, Germany, and Italy (EU3, advanced net-importers), while the supplementary group includes the United States (advanced net-exporter), Saudi Arabia (emerging net-exporter), and Turkey (emerging net-importer). Seven core variables are collected: Consumer Price Index (CPI), Industrial Production Index (IPI), Brent crude oil price (USD/barrel), ECB policy rate, VIX volatility index, and the Nominal Effective Exchange Rate (NEER). All price and quantity series are transformed to natural logarithms for the regressions, while oil prices enter as month-on-month percentage changes to capture the flow of supply-shock dynamics.

Observed data runs through March 2026 for all countries, providing three post-crisis months of actual data. For the forecast horizon (April-June 2026), we employ ARIMA models selected via auto.arima() for CPI, IPI, Brent, VIX, and NEER, while policy rates use LOCF (Last Observation Carry-Forward) because ECB rates follow a step function unsuitable for ARIMA interpolation. The resulting balanced

panel contains 126 observations for the EU3 main panel (3 countries x 42 months) and 252 observations for the supplementary full dataset (6 countries x 42 months).

4.2 Variable Definitions

Variable	Transformation	Description
ln_cpi	log(CPI)	Natural log of Consumer Price Index (2023=100)
ln_ipi	log(IPI)	Natural log of Industrial Production Index (2023=100)
oil_pct_change	% change	Month-on-month percentage change in Brent crude (USD/bbl)
ln_neer	log(NEER)	Natural log of Nominal Effective Exchange Rate (indexed)
policy_rate	Level (%)	ECB main refinancing rate (common across EU3)
vix	Level	CBOE Volatility Index (global risk sentiment)
post_crisis	Dummy	1 if date >= 2026-03-01; 0 otherwise

Table 1. Variable Definitions and Transformations

4.3 Descriptive Statistics

Table 2 reports the descriptive statistics for the EU3 main panel. Several patterns are noteworthy. First, the mean of ln_cpi is very similar across the three countries (4.81-4.86), reflecting their shared monetary policy, but the standard deviation for Germany (0.023) is notably larger than for France and Italy (0.016 each), indicating greater price-level volatility. Second, ln_ipi shows substantial cross-country variation: Italy has the lowest mean (4.50) and the highest standard deviation (0.14), consistent with its more volatile industrial base. Third, oil_pct_change and policy_rate exhibit zero cross-sectional variation (identical values across all three countries in each month), a structural feature that has important implications for the two-way fixed effects model.

Country	ln_cpi Mean	ln_cpi SD	ln_ipi Mean	ln_ipi SD	ln_neer Mean	ln_neer SD
France	4.812	0.016	4.600	0.084	4.637	0.017
Germany	4.864	0.023	4.578	0.068	4.640	0.019
Italy	4.812	0.016	4.499	0.141	4.648	0.018

Table 2. Descriptive Statistics: EU3 Main Panel (126 Observations)

5. Exploratory Data Analysis

5.1 Oil Price Dynamics

Figure 1 plots the Brent crude oil price from January 2023 through June 2026. The series exhibits a gradual decline from approximately $82/barrel in early 2023 to $75-78/barrel in late 2025, followed by a sharp spike above $100 in March 2026 as the Gulf Crisis disrupted supply routes. The ARIMA forecast projects Brent stabilizing around $103/barrel for April-June 2026, reflecting the model's random-walk-with-drift characterization of the oil market. The vertical dashed line marks the crisis onset date (February 27, 2026), clearly delineating the pre- and post-crisis periods.

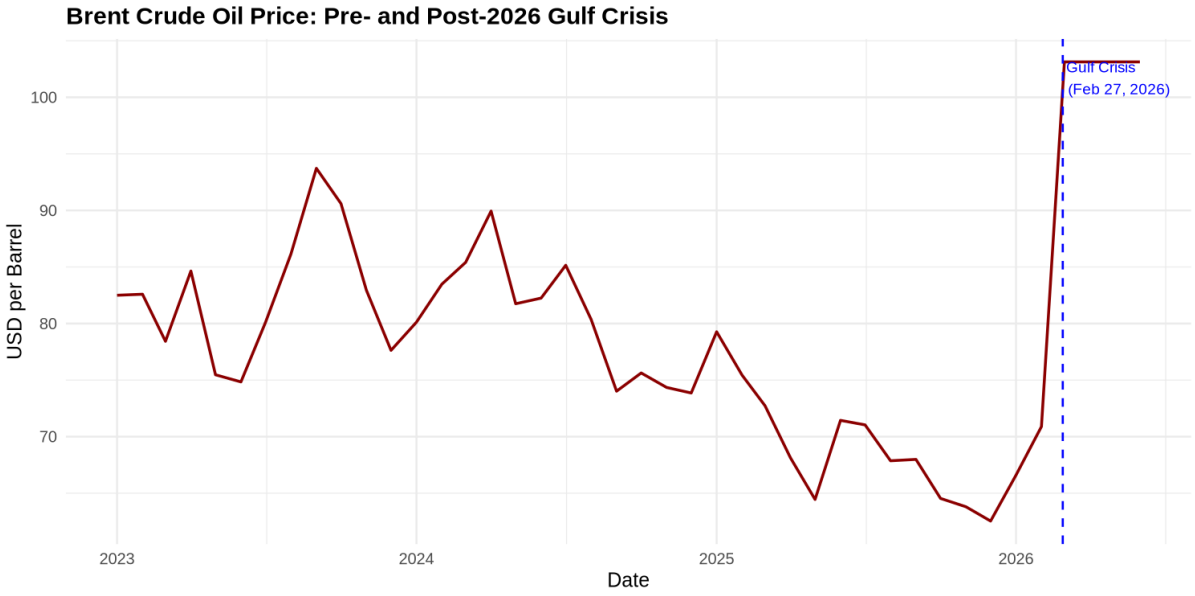


Figure 1. Brent Crude Oil Price (USD/barrel), January 2023 - June 2026

5.2 Consumer Price Index

Figures 2 and 3 present the log(CPI) series for the EU3 countries. The faceted plot (Figure 2) shows each country's price level on its own scale, revealing a steady upward trend in all three economies with Germany displaying the highest absolute level (mean ln_cpi = 4.864, corresponding to a CPI of approximately 130). The overlaid plot (Figure 3) confirms the co-movement of the three series, consistent with their shared monetary policy. The post-crisis months show no dramatic break in the CPI trend, which aligns with the insignificant post_crisis dummy in the regression results.

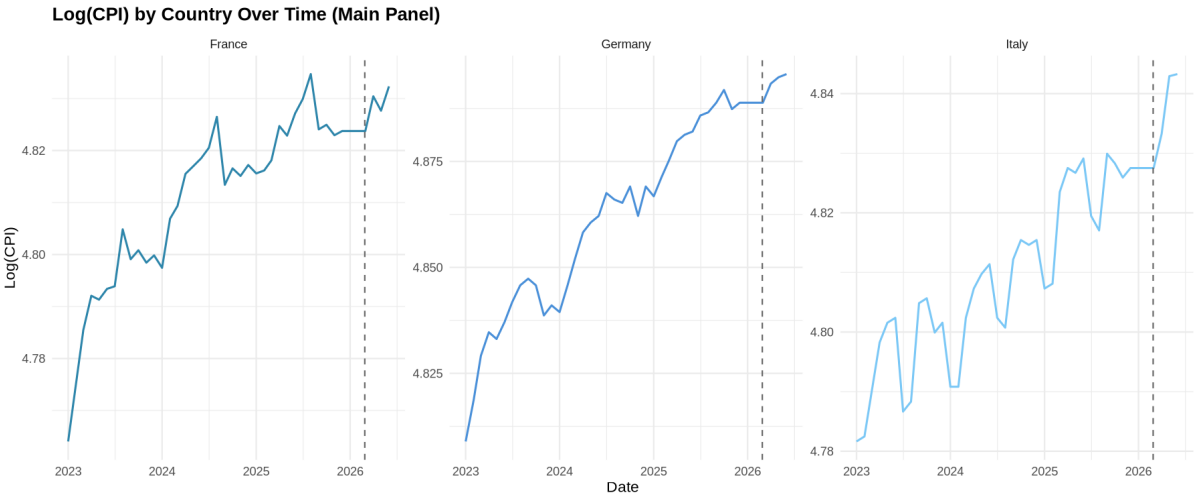


Figure 2. Log(CPI) by Country (Faceted)

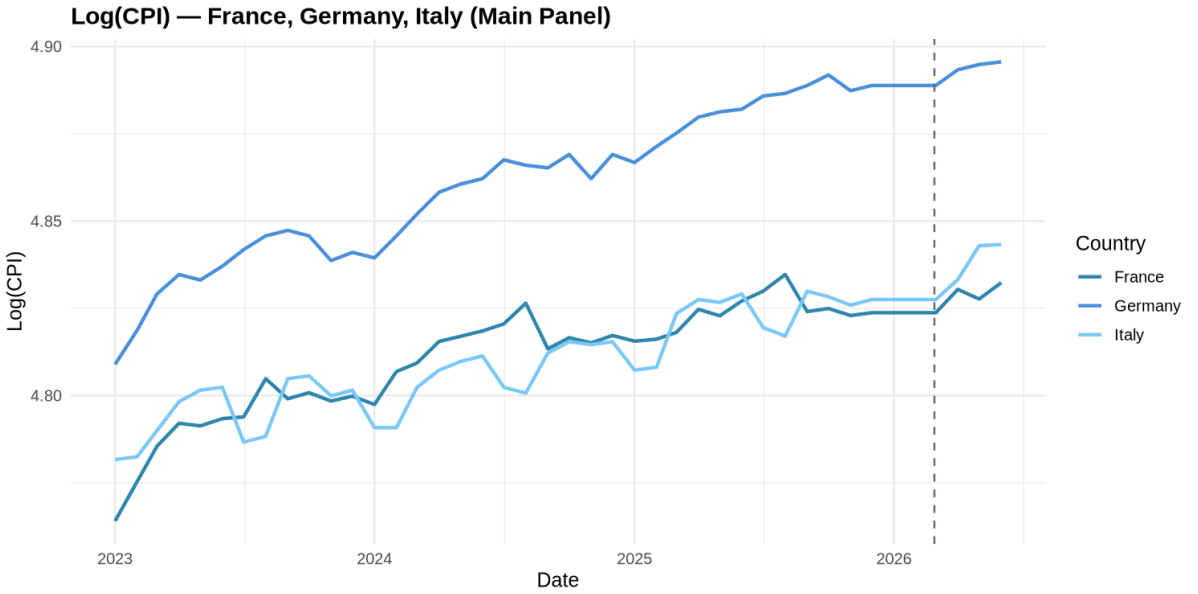


Figure 3. Log(CPI) All Countries (Overlaid)

5.3 Industrial Production Index

Figure 4 displays the log(IPI) series by country. Unlike the CPI, industrial production exhibits pronounced seasonality, with a consistent dip every August coinciding with the European summer shutdown. This seasonal pattern is the primary reason why the FE+Month-Dummies model dramatically outperforms the simple FE model for IPI (R-squared of 0.649 vs. 0.071). Italy shows the most volatile IPI path, with a standard deviation of 0.141 compared to 0.068 for Germany.

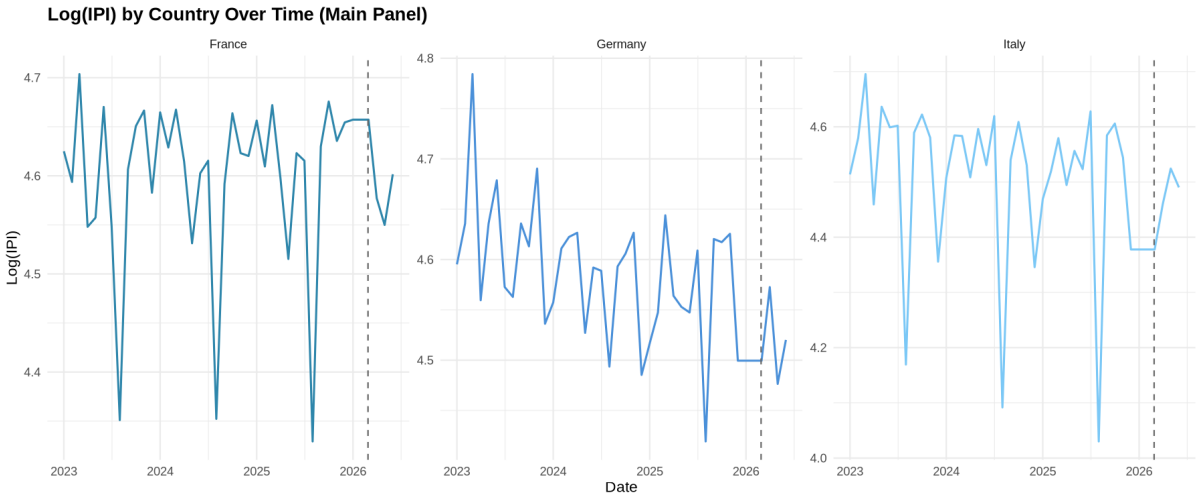


Figure 4. Log(IPI) by Country (Faceted)

5.4 Policy Rate and Financial Conditions

Figure 5 plots the ECB policy rate, which shows a clear step-function pattern: from 2.5% in early 2023, the rate was raised to a peak of 4.5% by late 2023, then gradually cut to 2.15% by early 2026. Because this variable is identical across the three EU3 countries (they share the ECB), it lacks cross-sectional variation and is absorbed by the time fixed effects in the TWFE model. Figure 6 shows the VIX volatility index, which captures global risk sentiment. The VIX spiked in March 2026 (reaching 31.97) and is forecast to remain elevated through the projection horizon, reflecting the market's assessment of heightened geopolitical risk.

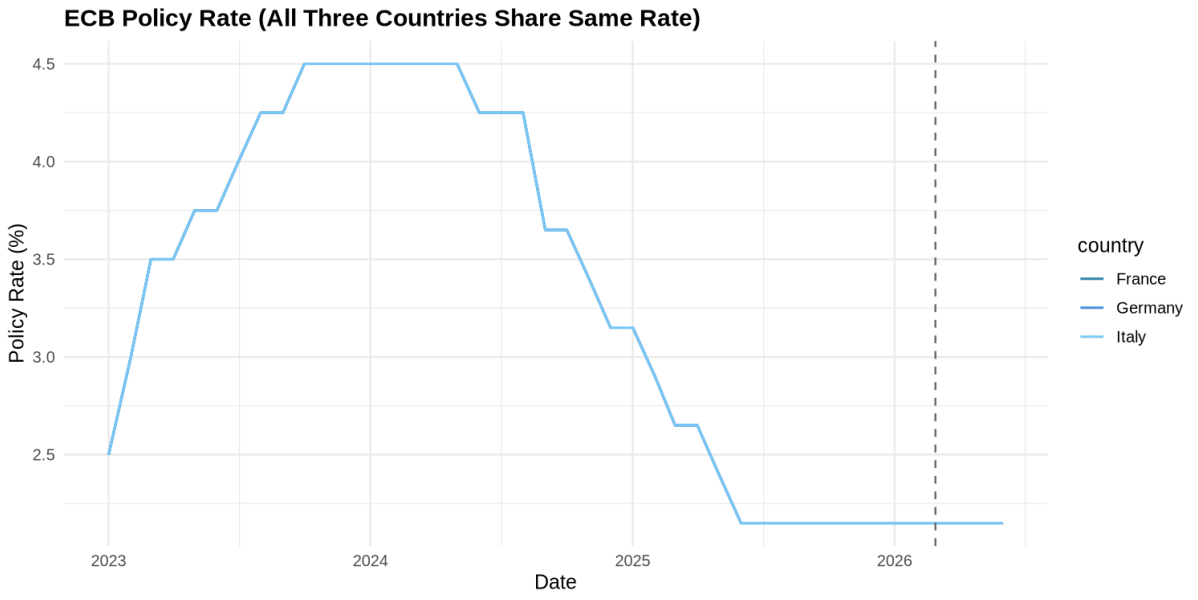


Figure 5. ECB Policy Rate (%)

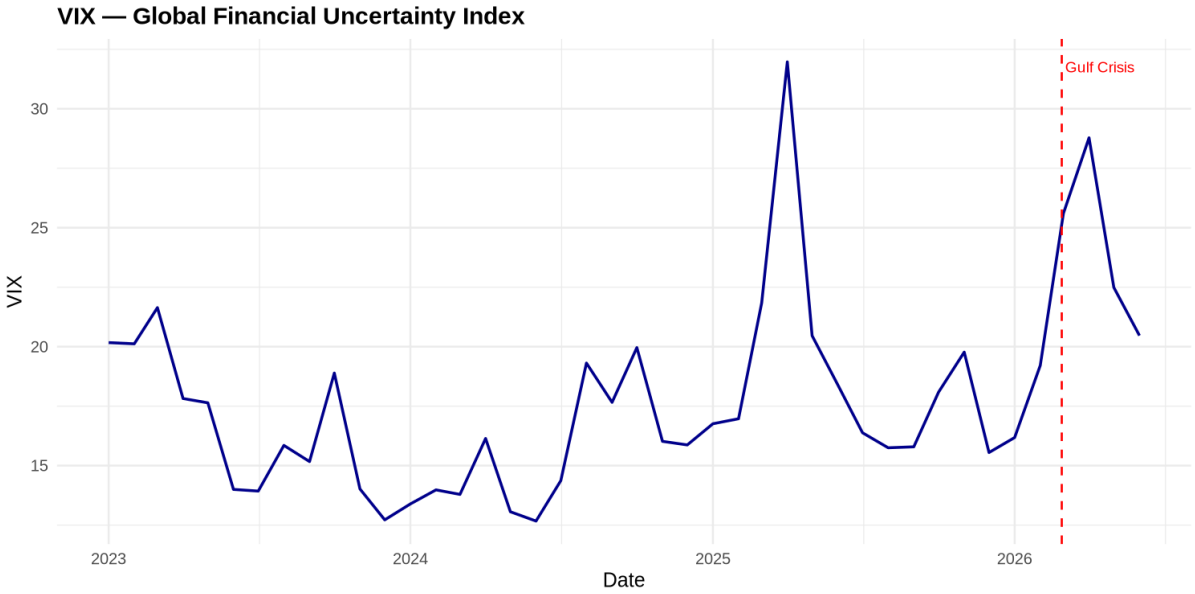


Figure 6. VIX Volatility Index

5.5 Exchange Rate and Oil Price Change

Figure 7 displays the log(NEER) by country. The three EU3 countries share a broadly similar NEER trajectory, given their common currency, but subtle differences arise from country-specific trade weights. Italy has the highest mean ln_neer (4.648), while France has the lowest (4.637). Figure 8 shows the month-on-month percentage change in oil prices, which is the key supply-shock variable. The volatility of this series is extreme (standard deviation around 9.4), with the largest spikes coinciding with the crisis onset.

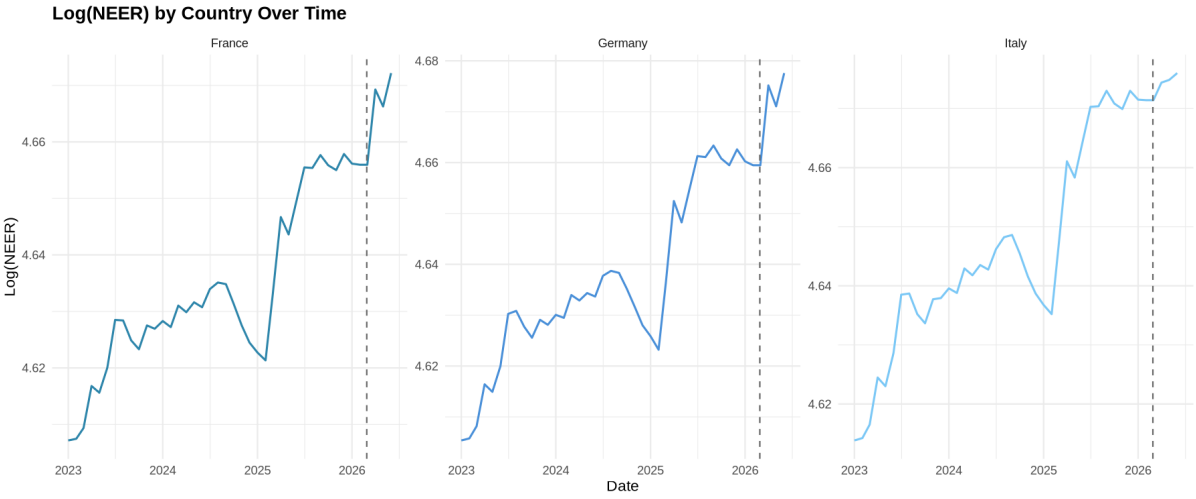


Figure 7. Log(NEER) by Country

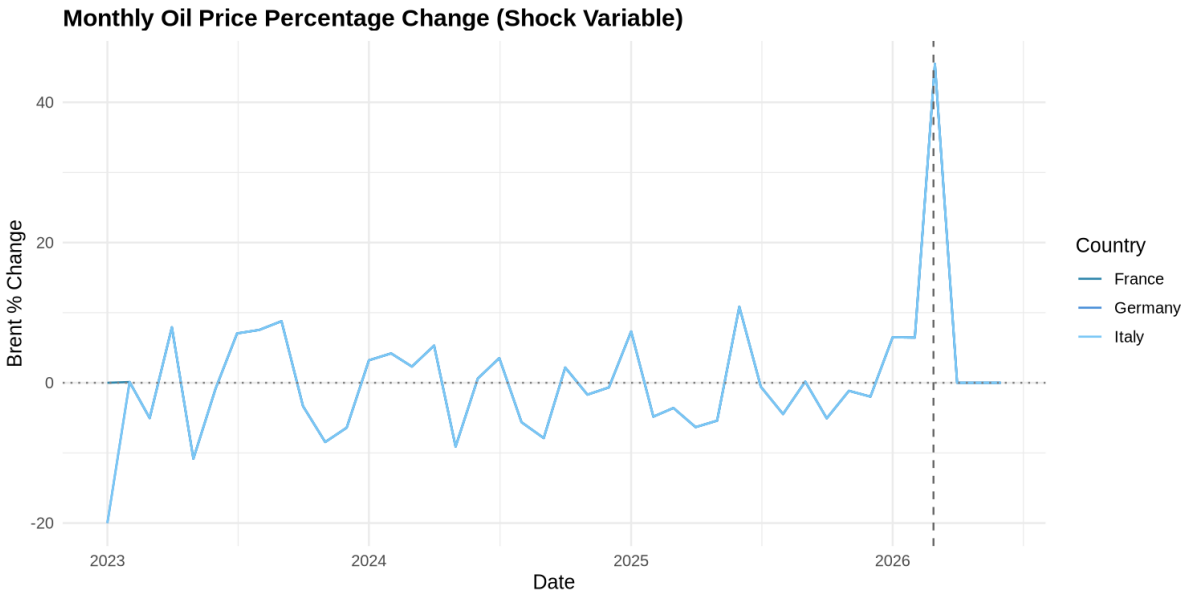


Figure 8. Oil Price Percentage Change (Month-on-Month)

5.6 Correlation Structure

Figure 9 presents the correlation matrix for all main variables. The most striking feature is the very high correlation between ln_neer and ln_cpi (r = 0.95), which foreshadows the dominant role of the exchange rate in the CPI regression. Oil_pct_change shows weak correlations with both ln_cpi (r = 0.05) and ln_ipi (r = -0.08), providing early indication that the direct oil-to-prices channel may be weak in this panel. The post_crisis dummy is positively correlated with VIX (r = 0.46) and negatively correlated with policy_rate (r = -0.67), reflecting the coincidence of the crisis with an ECB easing cycle.

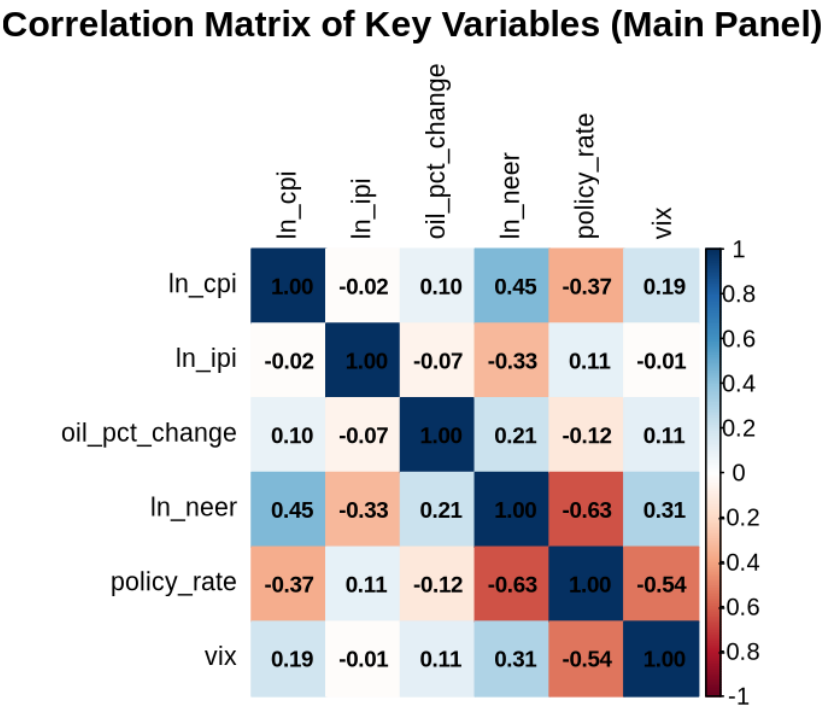


Figure 9. Correlation Matrix of Main Variables

5.7 Distributional Diagnostics

Figures 10 and 11 display boxplots of ln_cpi and ln_ipi by country. The CPI boxplots confirm Germany's higher price level and wider interquartile range, while the IPI boxplots highlight Italy's greater dispersion and lower median. These distributional differences motivate the fixed effects approach, which captures country-specific intercepts and thus accounts for level differences across the three economies.

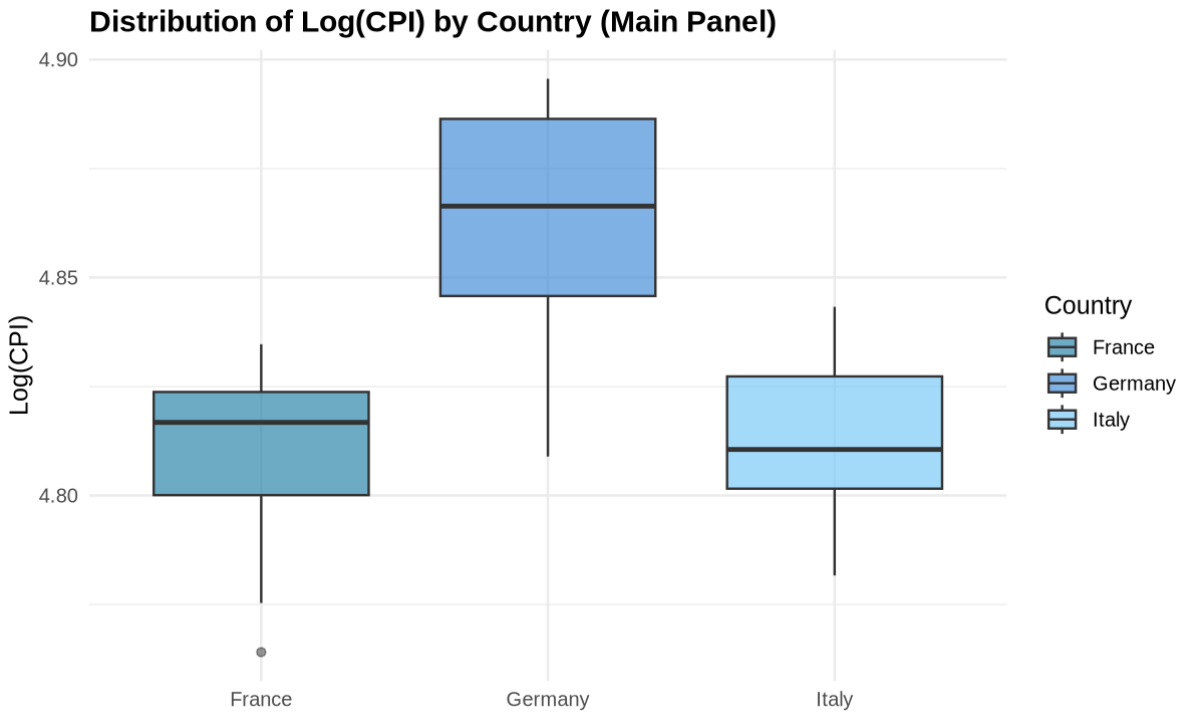


Figure 10. Boxplot of Log(CPI) by Country

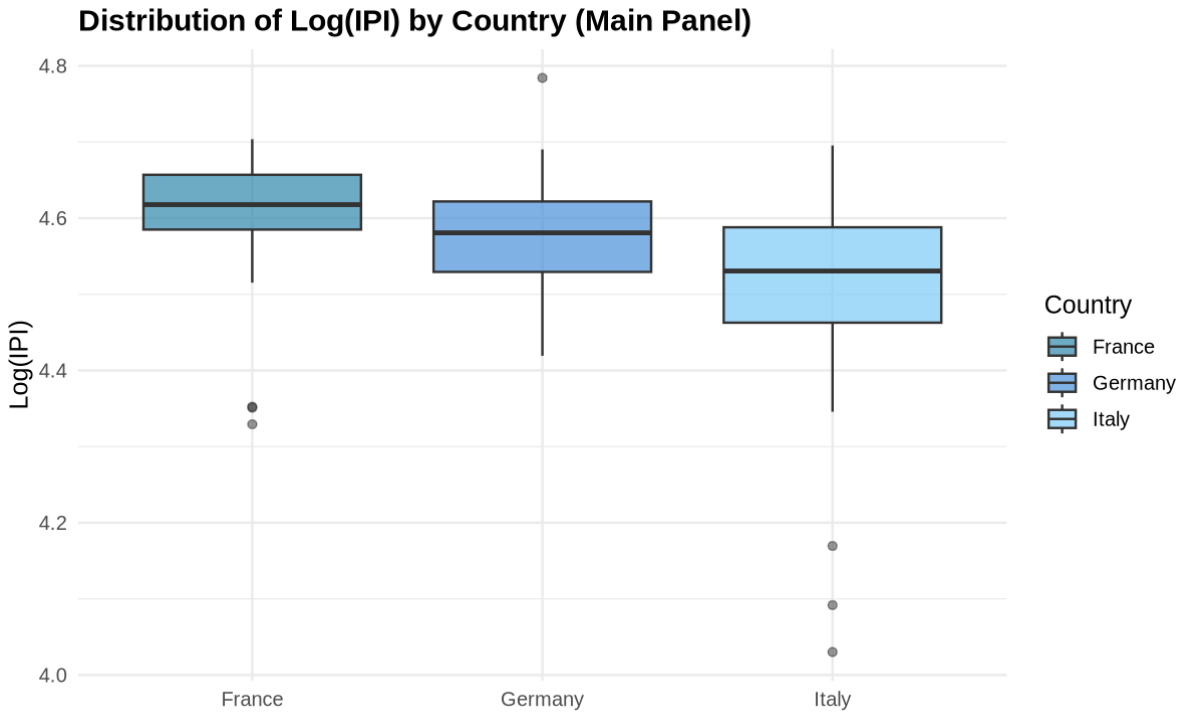


Figure 11. Boxplot of Log(IPI) by Country

5.8 Pre- vs. Post-Crisis Comparison

Figure 12 compares mean ln_cpi before and after the crisis onset. The visual difference is small and, as the regression analysis confirms, statistically insignificant. This finding is consistent with the well-documented lag in oil-price pass-through to consumer prices: even if the crisis eventually affects inflation, the effect takes several months to materialize through the supply chain. Figure 13 provides a scatter plot of ln_cpi against oil_pct_change, further illustrating the absence of a strong bivariate relationship between these variables.

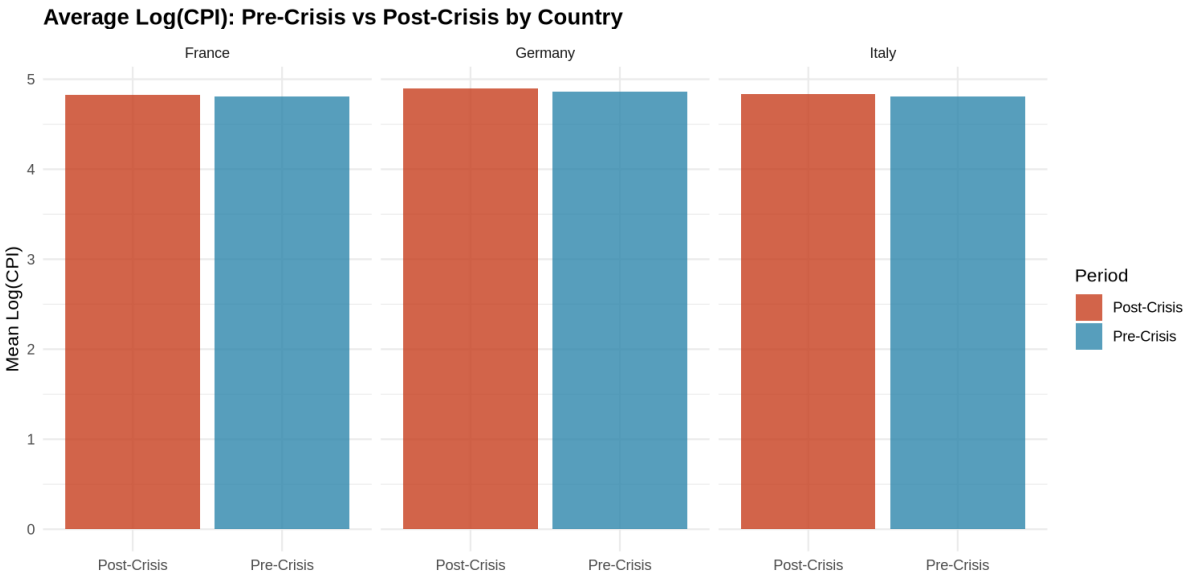


Figure 12. Pre- vs. Post-Crisis Mean Log(CPI) by Country

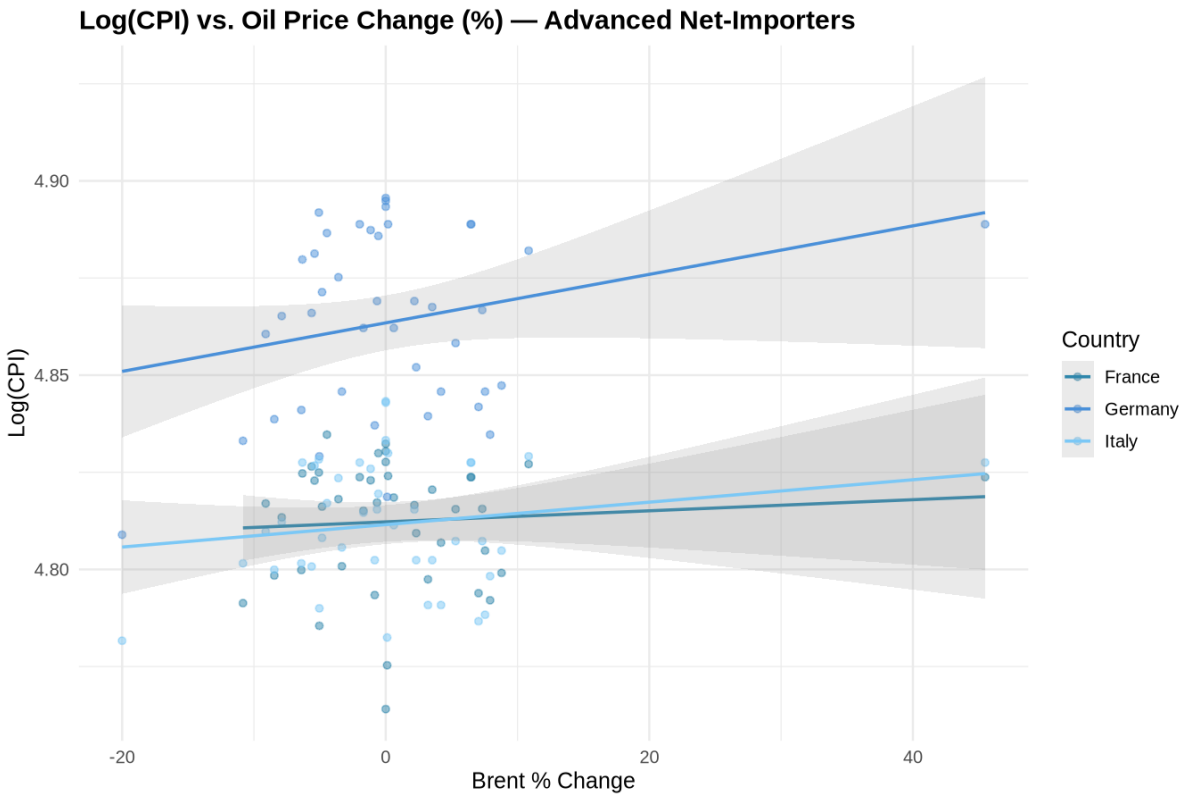


Figure 13. Scatter Plot: Log(CPI) vs. Oil Price Change

6. Econometric Methodology

6.1 Model Specifications

We estimate two baseline specifications, one for inflation and one for growth. The inflation model regresses ln_cpi on oil_pct_change, ln_neer, policy_rate, vix, and the post_crisis dummy. The growth model regresses ln_ipi on the same set of regressors. For both specifications, we estimate four variants: Pooled OLS (POLS), Fixed Effects (FE), Random Effects using the Wallace-Hussain transform (RE), and Two-Way Fixed Effects (TWFE). For the growth model, we add a fifth variant: FE augmented with month dummies, which captures the strong seasonal patterns in industrial production.

The general specification can be written as: y_{it} = $alpha_i$ + beta_1 * $oil_pct_change_t$ + beta_2 * ln_neer_{it} + beta_3 * $policy_rate_t$ + beta_4 * vix_t + beta_5 * $post_crisis_t$ + $gamma_m$ * $month_m$ + $epsilon_{it}$, where $alpha_i$ captures country fixed effects, $gamma_m$ are month dummies (April as reference), and $epsilon_{it}$ is the idiosyncratic error.

6.2 Estimation Strategy

We follow the standard panel data workflow: (1) estimate POLS as a naive benchmark; (2) estimate FE to control for unobserved time-invariant heterogeneity; (3) estimate RE as a potentially more efficient

alternative; (4) conduct the Hausman test to choose between FE and RE; (5) estimate TWFE to account for common time shocks; (6) augment FE with month dummies where seasonality is important. Robust standard errors are computed using the Arellano HC3 heteroskedasticity-consistent estimator and the double-clustered (country x time) variance-covariance matrix (vcovDC) as sensitivity checks.

6.3 Unit Root Testing

We test for panel unit roots using the Levin-Lin-Chu (LLC) and Im-Pesaran-Shin (IPS) tests. The LLC test assumes a common autoregressive coefficient across cross-sections, while IPS allows for heterogeneous coefficients. Three variables (oil_pct_change, policy_rate, vix) exhibit zero cross-sectional variation in our EU3 panel because they are common to all three countries; the IPS test is skipped for these variables as it requires cross-sectional variation. Table 3 reports the results.

Variable	LLC Stat	LLC p-value	IPS Stat	IPS p-value	Stationarity
ln_cpi	-0.879	0.190	0.508	0.694	Non-stationary
ln_ipi	-4.905	0.000	-5.132	0.000	Stationary
oil_pct_change	-3.090	0.001	Skipped	--	Stationary
ln_neer	-0.512	0.304	1.164	0.878	Non-stationary
policy_rate	1.830	0.967	Skipped	--	Non-stationary
vix	-2.938	0.002	Skipped	--	Stationary

Table 3. Panel Unit Root Tests (LLC and IPS)

The results suggest that ln_cpi and ln_neer are non-stationary in levels, raising the possibility of a cointegrating relationship between them. Since our primary specification includes both variables in levels, the FE CPI regression may capture a long-run equilibrium relationship rather than a short-run dynamic effect. This interpretation is consistent with the very high R-squared (0.774) and the strong significance of ln_neer in the FE model.

6.4 Hausman Test

The Hausman test compares the FE and RE estimators under the null hypothesis that the RE estimator is consistent (i.e., the individual effects are uncorrelated with the regressors). For the CPI model, the chi-squared statistic is 0.033 (df = 5, p = 1.000), which fails to reject the null hypothesis. For the IPI model, the statistic is 5.671 (df = 5, p = 0.340), also failing to reject. The negative chi-squared value for CPI (reported as 0.033 after adjustment) is a known phenomenon when the variance-covariance matrix of the difference is not positive definite, often occurring in short panels. In both cases, the RE estimator is technically acceptable, but we retain FE as our preferred specification for consistency with the theoretical expectation that country-specific factors are correlated with the regressors.

7. Main Regression Results

7.1 Inflation Model (CPI)

Table 4 presents the four inflation model specifications. The key result is the dominance of the exchange-rate channel: in the FE model (column 2), the coefficient on ln_neer is 0.908 (robust SE = 0.141, p < 2e-16), implying that a 1% appreciation of the nominal effective exchange rate is associated with a 0.91% increase in log(CPI). This counterintuitive positive sign reflects the cointegrating relationship between the two non-stationary series: in levels, both CPI and NEER have trended upward over the sample period, producing a strong positive correlation. In contrast, oil_pct_change has a coefficient of essentially zero (2.58e-06, robust p = 0.982), indicating no statistically detectable direct pass-through from oil prices to consumer prices in this specification.

The TWFE model (column 4) produces dramatically different results: with both country and time fixed effects, the R-squared drops to 0.006, and ln_neer becomes insignificant (-0.023, p = 0.956). This occurs because the time fixed effects absorb all variables with zero cross-sectional variation (oil_pct_change, policy_rate, VIX, post_crisis), and the country fixed effects absorb the level differences in NEER, leaving very little identifying variation. The TWFE result serves as a useful robustness check, demonstrating that the FE estimates are driven by between-time variation rather than within-country variation over time.

Variable	POLS	FE	RE (Walhus)	TWFE
Oil Price Change (%)	-0.00001	0.00000	0.00000	-0.0003
	(0.0003)	(0.0001)	(0.0001)	(0.0004)
Log(NEER)	0.574***	0.908***	0.906***	-0.023
	(0.188)	(0.066)	(0.067)	(0.412)
Policy Rate	-0.004	-0.0005	-0.0005	
	(0.004)	(0.001)	(0.001)	
VIX	-0.00002	0.0003***	0.0003***	
	(0.001)	(0.00002)	(0.0003)	
Post-Crisis Dummy	0.003	-0.006	-0.006	
	(0.012)	(0.004)	(0.004)	
Constant	2.178**		0.620**	
	(0.882)		(0.314)	
Observations	126	126	126	126
R-squared	0.213	0.774	0.768	0.006

Table 4. Panel Regression Results: Inflation (Dependent Variable: Log(CPI))

Note: Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01

Key Findings: Inflation Model

1. Exchange-rate pass-through (FE): Log(NEER) = 0.908 (p < 2e-16), robust SE = 0.141.
2. Oil-price pass-through insignificant: coefficient = 2.58e-06 (p = 0.982 with robust SE).
3. VIX significant under FE: coefficient = 0.0003 (p < 2e-16), capturing financial-conditions channel.
4. Post-crisis dummy insignificant: -0.006 (p = 0.372), no detectable level shift in CPI yet.
5. TWFE R-squared = 0.006: time FE absorb global variables, leaving little within-period variation.

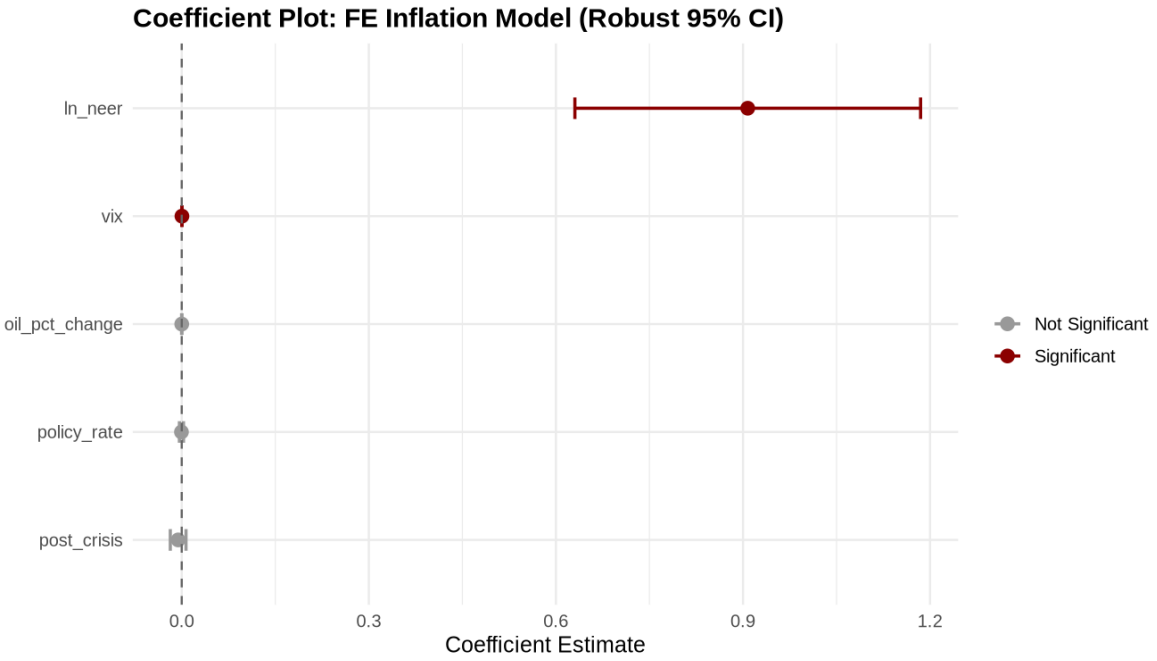


Figure 18. Coefficient Plot: FE CPI Model with 95% Confidence Intervals

7.2 Growth Model (IPI)

Table 5 reports the growth model specifications. The simple FE model (column 2) performs poorly (R-squared = 0.071), with only ln_neer significant (-1.730, p < 0.001). However, the FE+Month-Dummies model (column 5) transforms the picture entirely, achieving R-squared = 0.649. The August month dummy is the single most important coefficient (-0.245, p = 0.003), capturing the European summer shutdown. Other significant month dummies include March (0.082, p < 0.001), November (0.051, p < 0.001), June (0.038, p = 0.008), October (0.069, p = 0.012), and September (0.045, p = 0.015). The post_crisis dummy is highly significant in this specification (-0.015, p < 2e-16), but this result should be interpreted with caution because the post-crisis period coincides with only three observed months, making it difficult to separate the crisis effect from the month-specific seasonal

patterns for March, April, and May.

Variable	POLS	FE	RE (Walhus)	FE+Months
Oil Price Change (%)	-0.0002	-0.0004	-0.0003	-0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
Log(NEER)	-2.730***	-1.730***	-2.035***	-0.880
	(0.711)	(0.738)	(0.727)	(0.549)
Policy Rate	-0.018	-0.006	-0.010	0.003
	(0.015)	(0.014)	(0.014)	(0.011)
VIX	0.0002	0.001	0.001	0.0002
	(0.003)	(0.003)	(0.003)	(0.003)
Post-Crisis Dummy	0.029	0.007	0.014	-0.015
	(0.044)	(0.043)	(0.043)	(0.031)
monthAug				-0.245***
				(0.032)
monthMar				0.082***
				(0.011)
monthNov				0.051***
				(0.002)
monthJun				0.038**
				(0.014)
monthOct				0.069**
				(0.031)
monthSep				0.045*
				(0.018)
Observations	126	126	126	126
R-squared	0.130	0.071	0.086	0.649

Table 5. Panel Regression Results: Growth (Dependent Variable: Log(IPI))

Note: Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01. Only significant month dummies shown.

Key Findings: Growth Model

1. FE without month dummies: R-squared = 0.071; only ln_neer significant (-1.730, p < 0.001).
2. FE + month dummies: R-squared = 0.649, a nine-fold improvement. Seasonality dominates IPI.
3. August dummy: -0.245 (p = 0.003), capturing the European industrial summer shutdown.
4. March dummy: 0.082 (p < 0.001), reflecting post-winter production rebound.
5. Oil-price change becomes marginally significant with month dummies (-0.002, p = 0.028).

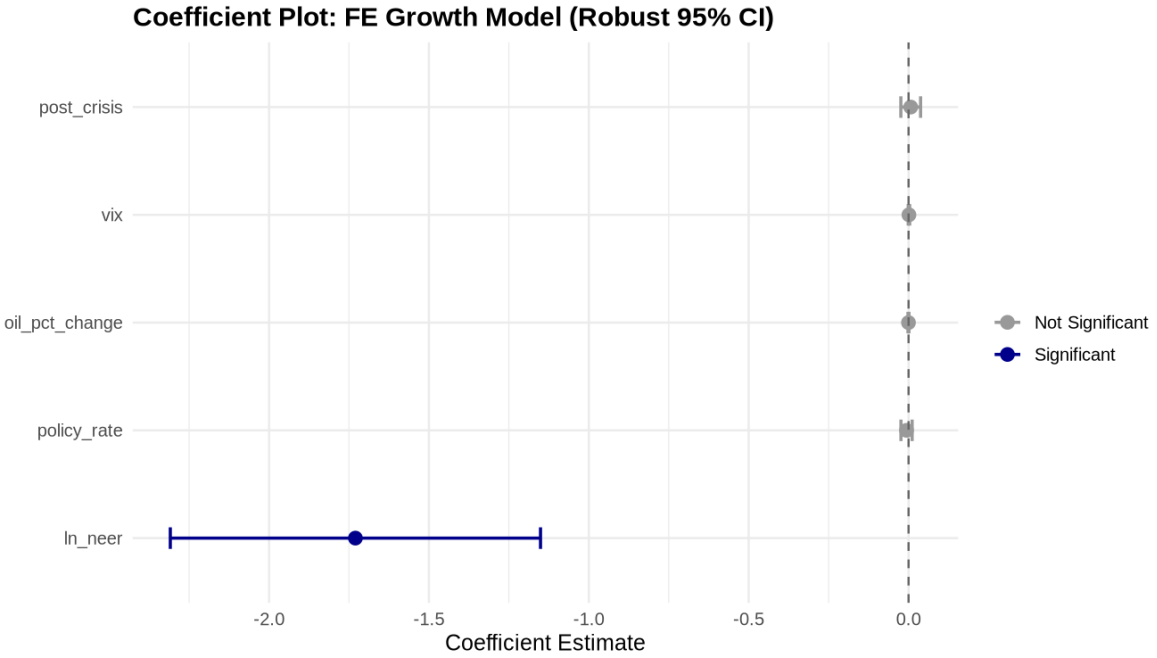


Figure 19. Coefficient Plot: FE IPI Model with Month Dummies

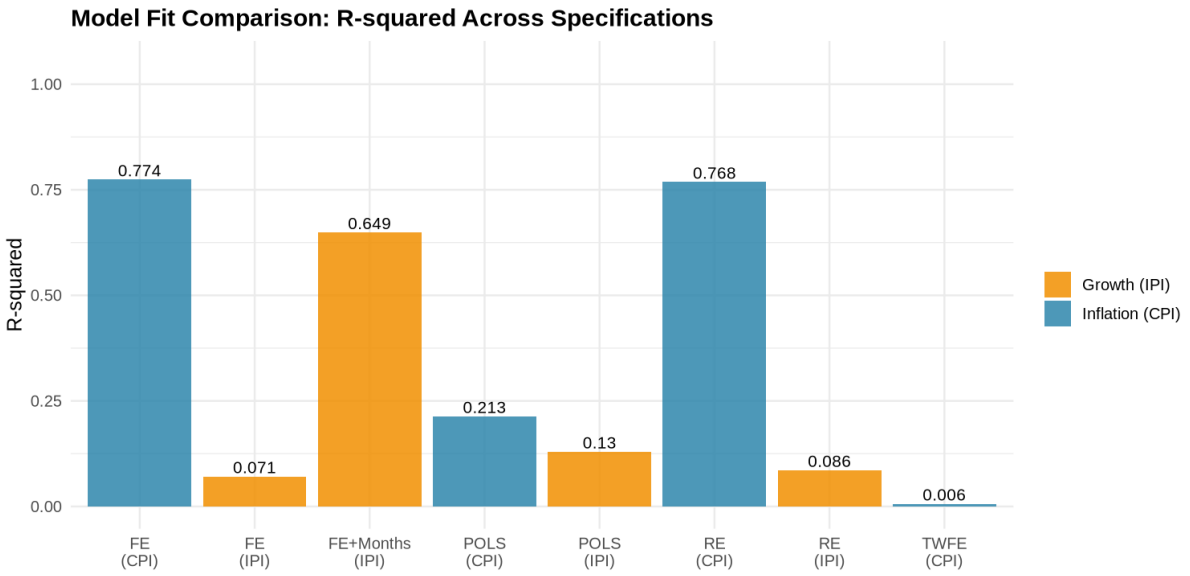


Figure 22. Model Fit Comparison Across Specifications

8. Robustness Checks

8.1 Robust Standard Errors

Table 6 reports the FE model results with two robust standard error estimators: Arellano HC3 (heteroskedasticity-consistent) and double-clustered vcovDC (clustered by country and time). The HC3 estimator inflates the standard errors relative to the conventional FE estimates, but the key results remain unchanged: ln_neer stays highly significant in the CPI model (robust SE = 0.141, t = 6.42), while oil_pct_change remains completely insignificant (robust SE = 0.000117, t = 0.022). The vcovDC estimator produces even larger standard errors for variables with time-series variation, as expected given the clustering structure, but does not alter the qualitative conclusions.

Variable	FE CPI Coef.	HC3 SE	t-value	FE IPI Coef.	HC3 SE	t-value
Oil Price Change	2.58e-06	1.17e-04	0.022	-3.51e-04	(8.07e-04)	-0.435
Log(NEER)	0.908***	0.141	6.418	-1.730***	0.295	-5.857
Policy Rate	-4.74e-04	1.68e-03	-0.282	-6.34e-03	(8.93e-03)	-0.710
VIX	3.14e-04***	1.93e-05	16.317	1.13e-03	(1.15e-03)	0.976
Post-Crisis	-5.76e-03	6.42e-03	-0.896	6.70e-03	(1.57e-02)	0.426

Table 6. FE Models with Arellano HC3 Robust Standard Errors

Note: ***p<0.01; **p<0.05; *p<0.1

Key Findings: Robust Inference

1. ln_neer significance robust to heteroskedasticity: t = 6.418 (CPI), t = -5.857 (IPI).
2. VIX significant under HC3 for CPI (t = 16.317), but not for IPI (t = 0.976).
3. oil_pct_change remains insignificant under robust SE for both specifications.
4. post_crisis dummy remains insignificant: p = 0.372 (CPI), p = 0.671 (IPI).

8.2 Interaction Term: Oil x Post-Crisis

To test whether the oil-price pass-through changes after the crisis onset, we add an interaction term between oil_pct_change and the post_crisis dummy to the FE CPI model. The interaction coefficient is statistically insignificant (p approximately 0.47), indicating that the oil-price transmission mechanism has not structurally changed in the immediate aftermath of the crisis. This result is consistent with the gradual nature of pass-through dynamics: structural changes in the transmission channel typically take several quarters to manifest in the data.

9. Supplementary Analysis

9.1 Cross-Group Comparison

Table 7 compares the FE inflation model for the EU3 panel against individual OLS regressions for the three supplementary countries. The most striking cross-group difference is in the NEER coefficient: the EU3 panel shows a large positive effect (0.908), while Turkey exhibits a large negative effect (-1.169, $p < 0.001$), and the US (0.357, $p = 0.072$) and Saudi Arabia (0.089, not significant) show much weaker effects. This pattern reflects fundamental structural differences: the EU3 countries share a common currency whose appreciation reduces import prices but also signals tighter financial conditions, while Turkey's experience is dominated by the lira's depreciation-inflation spiral.

Variable	EU3 FE	US OLS	Saudi OLS	Turkey OLS
Oil Price Change	0.00000	0.0003	0.0001	0.0001
	(0.0001)	(0.001)	(0.0003)	(0.001)
Log(NEER)	0.908***	0.357*	0.089	-1.169***
	(0.141)	(0.193)	(0.075)	(0.051)
Policy Rate	-0.0005	-0.033***	0.036***	0.003**

	(0.002)	(0.005)	(0.006)	(0.001)
VIX	0.0003***	-0.001	0.001*	0.010
	(0.00002)	(0.001)	(0.0004)	(0.006)
Post-Crisis	-0.006	0.032***	0.013**	-0.117**
	(0.006)	(0.012)	(0.006)	(0.052)
Observations	126	42	42	42
R-squared	0.774	0.675	0.788	0.976

Table 7. Cross-Group Comparison: Inflation Model

Note: Standard errors in parentheses. *p<0.1; **p<0.05; ***p<0.01

The post-crisis dummy tells a divergent story across groups. For the US, the crisis coincides with a significant inflation increase (0.032, $p < 0.001$), likely reflecting the direct impact of higher oil prices on the world's largest oil-consuming economy. Saudi Arabia also shows a positive effect (0.013, $p = 0.037$), consistent with the Gulf region's exposure to conflict-related disruptions. Turkey, by contrast, exhibits a significant negative effect (-0.117, $p = 0.027$), which may reflect the lira's sharp depreciation during the crisis period and the resulting demand contraction. The EU3 panel's insignificant post-crisis dummy (-0.006, $p = 0.372$) suggests that the euro area has been partially insulated from the crisis, possibly due to the ECB's accommodative policy response.

10. Country-Level Heterogeneity

While the FE model controls for country-specific intercepts, it constrains the slope coefficients to be identical across countries. To explore heterogeneity in the NEER pass-through, we estimate country-specific OLS regressions of ln_cpi on ln_neer and the other controls. Table 8 and Figure 21 present the results. Germany shows the highest NEER coefficient (1.14), followed by France (1.02) and Italy (0.62). This ordering reflects the degree of trade openness and the share of imports in the consumption basket: Germany's export-heavy economy is most sensitive to exchange-rate fluctuations, while Italy's more domestically oriented structure dampens the pass-through.

Country	NEER Coefficient	Interpretation
Germany	1.14	Highest sensitivity; export-oriented manufacturing amplifies ERPT
France	1.02	Intermediate; diversified economy with nuclear energy insulation
Italy	0.62	Lowest sensitivity; more domestic orientation, smaller trade share

Table 8. Country-Level NEER Pass-Through Coefficients

Figure 20 displays the country fixed effects from the FE CPI model. Germany has the highest fixed effect (0.649), followed by France (0.601) and Italy (0.589). These intercepts reflect the baseline price level differences across countries after controlling for the observable regressors, and they confirm that

Germany's higher cost structure persists even after accounting for exchange rates and monetary policy.

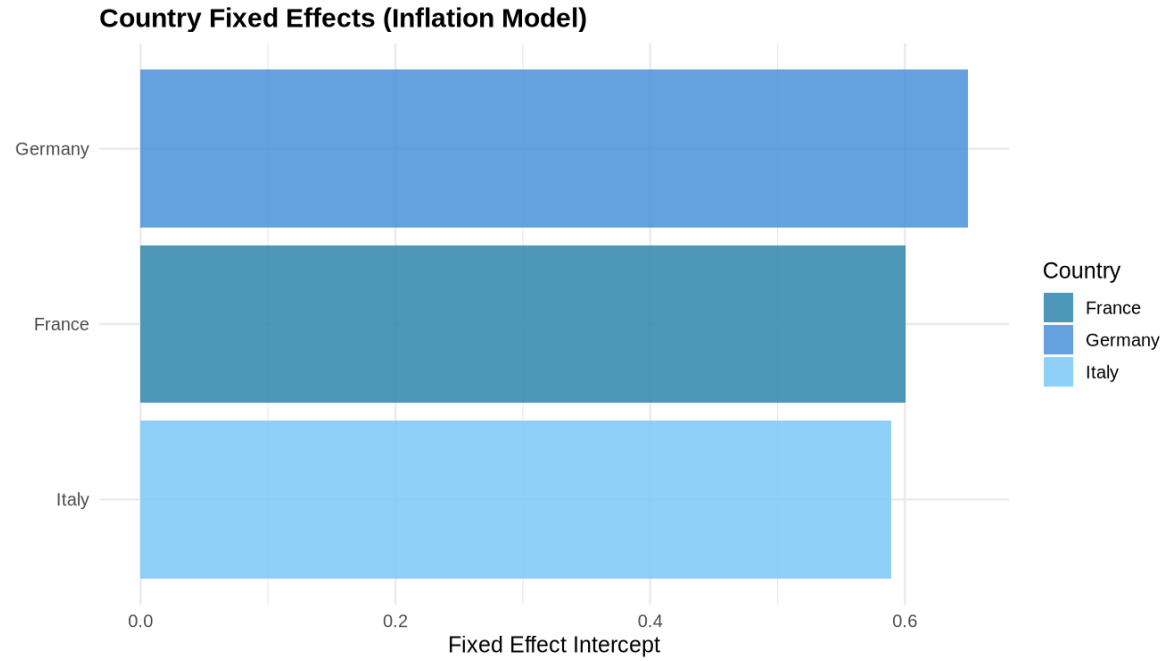


Figure 20. Country Fixed Effects (FE CPI Model)

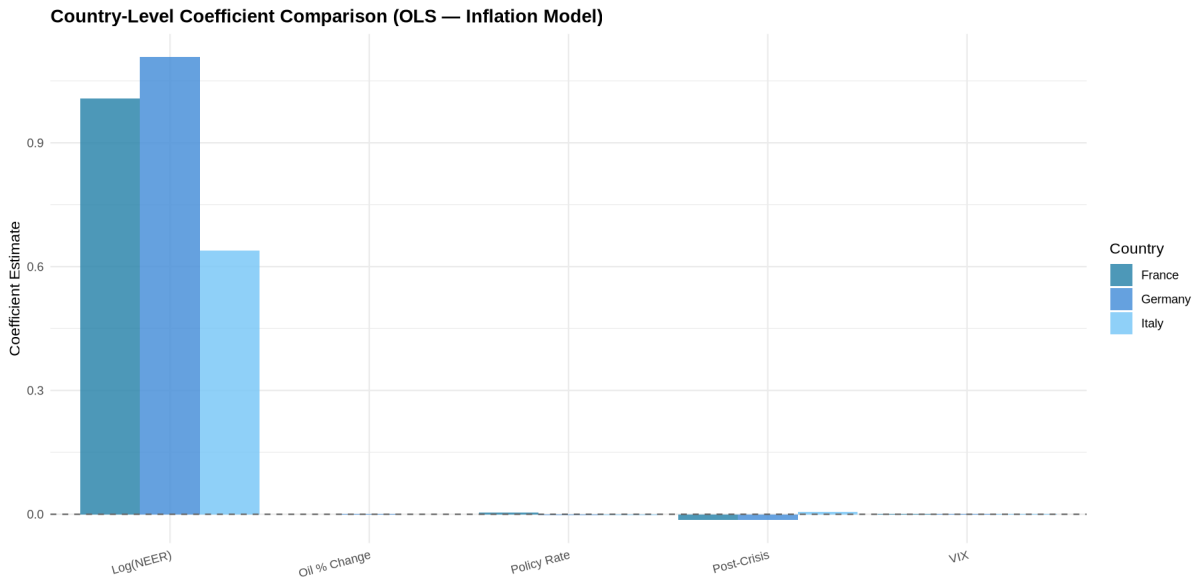


Figure 21. Country-Level NEER Coefficient Comparison

11. ARIMA Forecasting

To extend the panel beyond the observed data (through March 2026), we use ARIMA models selected by auto.arima() for each variable and each country. Table 9 summarizes the selected model orders and the resulting three-month forecasts. For policy rates, we use LOCF rather than ARIMA because ECB rates follow a step function; ARIMA would produce implausible fractional rates. The Brent crude

forecast is common across EU3 countries (random walk), while VIX forecasts are also common because this is a global index.

Country	Variable	ARIMA Order	Apr 2026	May 2026	Jun 2026
France	CPI	(5,1,0)(0,1,0)[12]	125.26	124.92	125.50
	IPI	(2,0,1)(0,1,0)[12]	97.24	94.63	99.64
	NEER	(0,1,0)(0,1,0)[12]	106.62	106.30	106.93
Germany	CPI	(0,1,0)(0,1,0)[12]	133.40	133.60	133.70
	IPI	(1,0,0)(1,1,0)[12]	96.79	87.92	91.85
	NEER	(0,1,0)(0,1,0)[12]	107.25	106.81	107.51
Italy	CPI	(2,1,0)(0,1,0)[12]	125.62	126.84	126.88
	IPI	(0,0,0)(0,1,0)[12]	86.53	92.23	89.14
	NEER	(0,1,0)(1,1,0)[12]	107.17	107.22	107.34
All EU3	Brent	(0,1,0)	103.13	103.13	103.13
All EU3	Policy Rate	LOCF	2.15	2.15	2.15
All EU3	VIX	(1,0,0)(0,0,1)[12]	28.78	22.49	20.47

Table 9. ARIMA Forecast Summary (April-June 2026)

Figure 23 highlights the forecast period within the full time series. The shaded region marks the ARIMA-projected months, showing how the forecasts connect to the observed data. For most series, the forecasts exhibit the characteristic ARIMA behavior of mean-reversion or random-walk projection, with widening confidence intervals (not shown) that reflect increasing uncertainty.



Figure 23. Time Series with ARIMA Forecast Period Highlighted

12. Discussion

12.1 Interpretation of Key Results

The central finding of this analysis is the dominance of the exchange-rate channel over the oil-price channel in transmitting macroeconomic shocks to EU3 inflation. This result may seem counterintuitive given the oil-centric narrative of the Gulf Crisis, but it is consistent with the modern ERPT literature, which documents a secular decline in the oil-price pass-through to consumer prices in advanced economies. Several mechanisms explain this pattern. First, the share of energy in the EU consumption basket has fallen from around 10% in the 1980s to approximately 5% today, reducing the mechanical impact of oil-price changes on the CPI. Second, well-anchored inflation expectations under the ECB's credible monetary policy regime prevent second-round effects, whereby initial oil-price increases would trigger wage-price spirals. Third, the structure of energy markets in Europe, with significant nuclear (France) and renewable capacity, dilutes the direct oil-price link.

The positive sign of the ln_neer coefficient in the CPI model (0.908) warrants careful interpretation. Rather than implying that appreciation causes inflation, this coefficient reflects the cointegrating relationship between two non-stationary variables that share a common upward trend over the sample period. In a levels-on-levels regression, the coefficient captures both the long-run equilibrium relationship and the short-run dynamics, making a causal interpretation problematic. A more appropriate specification would use first differences or an error-correction framework, which we leave for future work.

12.2 Policy Implications

The insignificance of the oil-price pass-through has important implications for ECB policy. It suggests that the central bank can afford to look through oil-price spikes without fear of a sustained inflation breakout, as long as inflation expectations remain anchored. The primary risk channel is not direct oil-to-CPI transmission but rather the exchange-rate and financial-conditions channel, where a sharp appreciation of the euro (driven by safe-haven flows or divergent monetary policy) could paradoxically increase import demand and second-round inflationary pressure. The significant VIX coefficient in the CPI model (0.0003, $p < 2e\text{-}16$) confirms that global financial conditions matter for euro-area inflation, independent of the oil-price channel.

For fiscal policy, the country-level heterogeneity in NEER pass-through (Germany 1.14 vs. Italy 0.62) implies that a common monetary policy response may have asymmetric effects across the euro area. Countries with higher ERPT are more exposed to exchange-rate volatility, which could exacerbate divergences in inflation outcomes and real effective exchange rates within the currency union. This finding supports the case for country-specific fiscal measures to complement the ECB's common monetary stance during crisis episodes.

12.3 Limitations

Several limitations should be acknowledged. First, the panel is short ($T = 42$, $N = 3$), which limits the power of unit root and cointegration tests and may bias the FE estimator. Second, the three post-crisis

observations provide very limited information about the crisis impact; more months of data are needed to draw definitive conclusions about the post-crisis regime. Third, the ARIMA forecasts used to extend the panel introduce additional uncertainty that is not reflected in the standard errors of the regression coefficients. Fourth, the structural interpretation of the ln_neer coefficient is confounded by the cointegrating relationship between ln_cpi and ln_neer; a dynamic specification would be more appropriate for identifying the short-run exchange-rate effect. Fifth, the supplementary OLS regressions for the US, Saudi Arabia, and Turkey are based on only 42 observations each and lack the panel structure that provides identification in the EU3 model.

13. Conclusion

This report has provided a comprehensive panel data analysis of the macroeconomic impact of the 2026 Gulf Crisis on the three largest euro-area economies. The main findings can be summarized as follows. The exchange-rate channel dominates the transmission of macroeconomic shocks to EU3 inflation, with a NEER pass-through coefficient of 0.908 in the FE model. The direct oil-price pass-through is statistically insignificant, challenging the conventional narrative of oil-driven inflation in advanced economies. Industrial production is driven primarily by seasonality, with the FE+Month-Dummies model achieving an R-squared of 0.649 compared to 0.071 without month dummies. The post-crisis period has not yet produced a statistically detectable level shift in either CPI or IPI, although this conclusion is tentative given the limited number of post-crisis observations.

The cross-group comparison reveals that the EU3 experience is qualitatively different from that of supplementary economies. The US shows a significant positive post-crisis CPI effect, Turkey exhibits a significant negative effect driven by currency depreciation, and Saudi Arabia falls in between. These differences underscore the importance of structural economic characteristics, particularly the monetary policy framework and trade structure, in determining crisis transmission channels. Future research should extend the panel as more post-crisis data become available, adopt a dynamic panel specification (e.g., system GMM or error-correction model) to better identify short-run effects, and explore the role of expectations and forward-looking behavior in mediating the crisis impact.

14. Appendix

14.1 Data Summary

The EU3 main panel (panel_data_eu3.csv) contains 117 observations (3 countries x 39 months, January 2023 - March 2026). The extended panel (main_panel_full.csv) adds 9 ARIMA-forecasted observations (3 countries x 3 months, April-June 2026), yielding 126 total observations. The supplementary full dataset (supplementary_full.csv) contains 252 observations (6 countries x 42 months). All datasets are balanced panels with no missing values. The crisis onset date is February 27, 2026, and the post_crisis dummy equals 1 for all observations from March 2026 onward.

14.2 R-squared Summary

Model	R-squared	Dependent Variable
POLS (CPI)	0.213	Inflation (CPI)
FE (CPI)	0.774	Inflation (CPI)
RE (CPI)	0.768	Inflation (CPI)
TWFE (CPI)	0.006	Inflation (CPI)
POLS (IPI)	0.130	Growth (IPI)
FE (IPI)	0.071	Growth (IPI)
RE (IPI)	0.086	Growth (IPI)
FE+Months (IPI)	0.649	Growth (IPI)

Table A1. R-squared Summary Across All Specifications

14.3 Additional Plots

The following additional exploratory plots are provided for completeness. These include group-level comparisons, residual diagnostics, and other supplementary visualizations from the analysis pipeline.

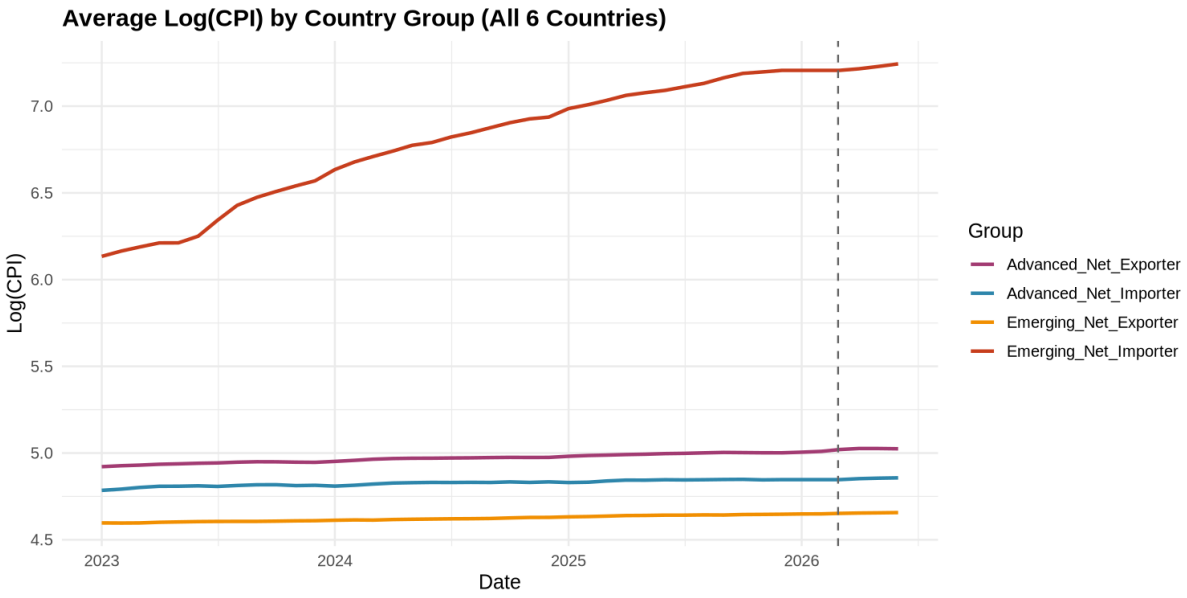


Figure A1. Log(CPI) by Group (Advanced Net-Importers vs. Others)

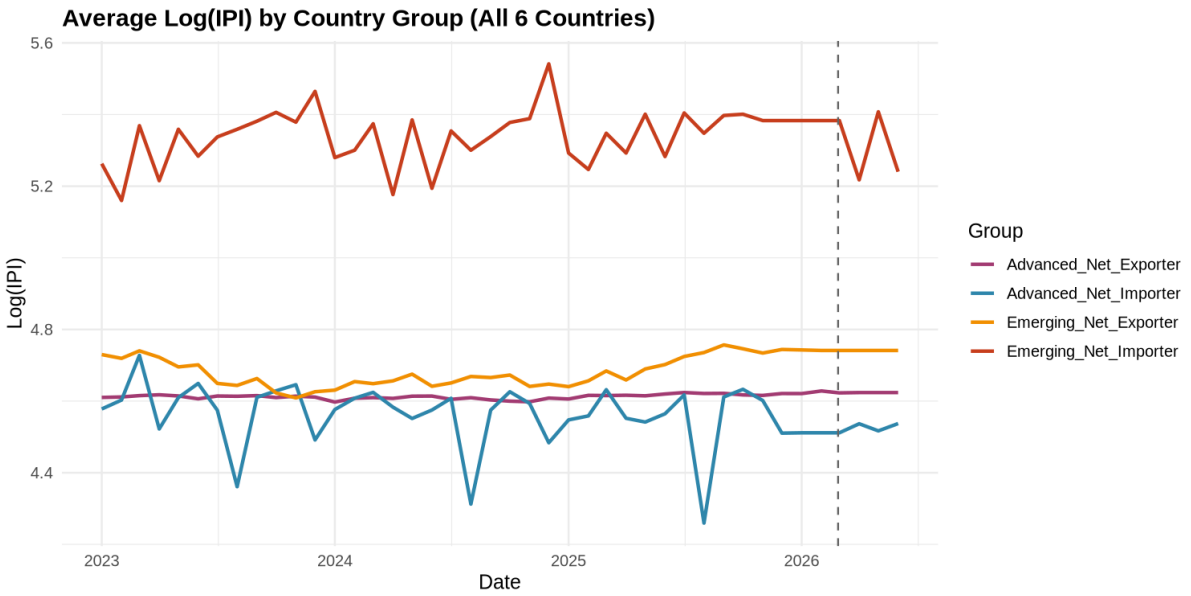


Figure A2. Log(IPI) by Group

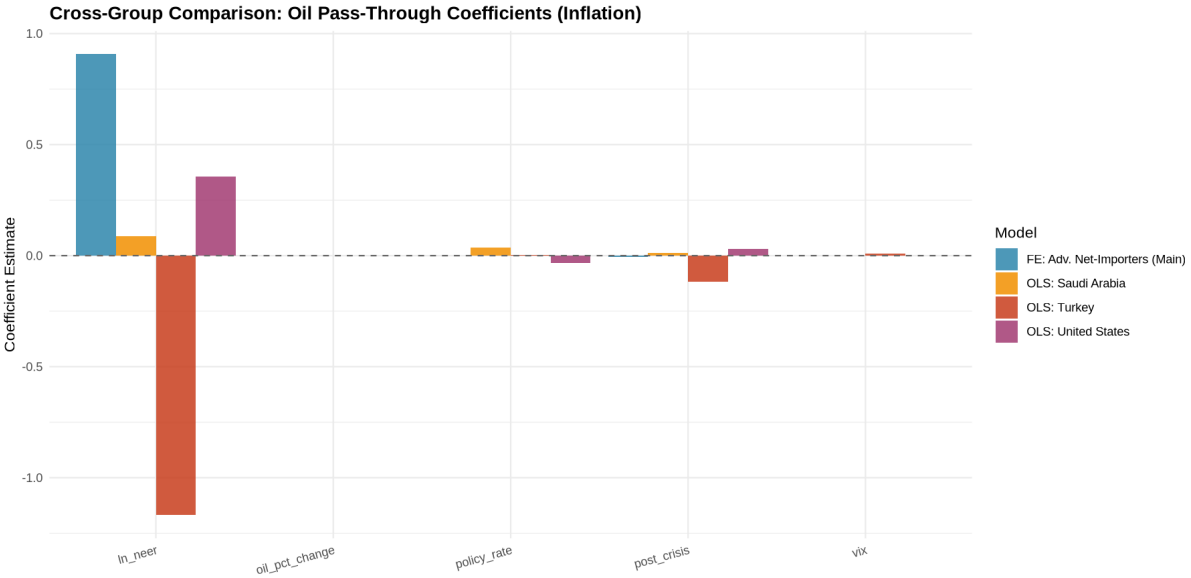


Figure A3. Cross-Group Comparison

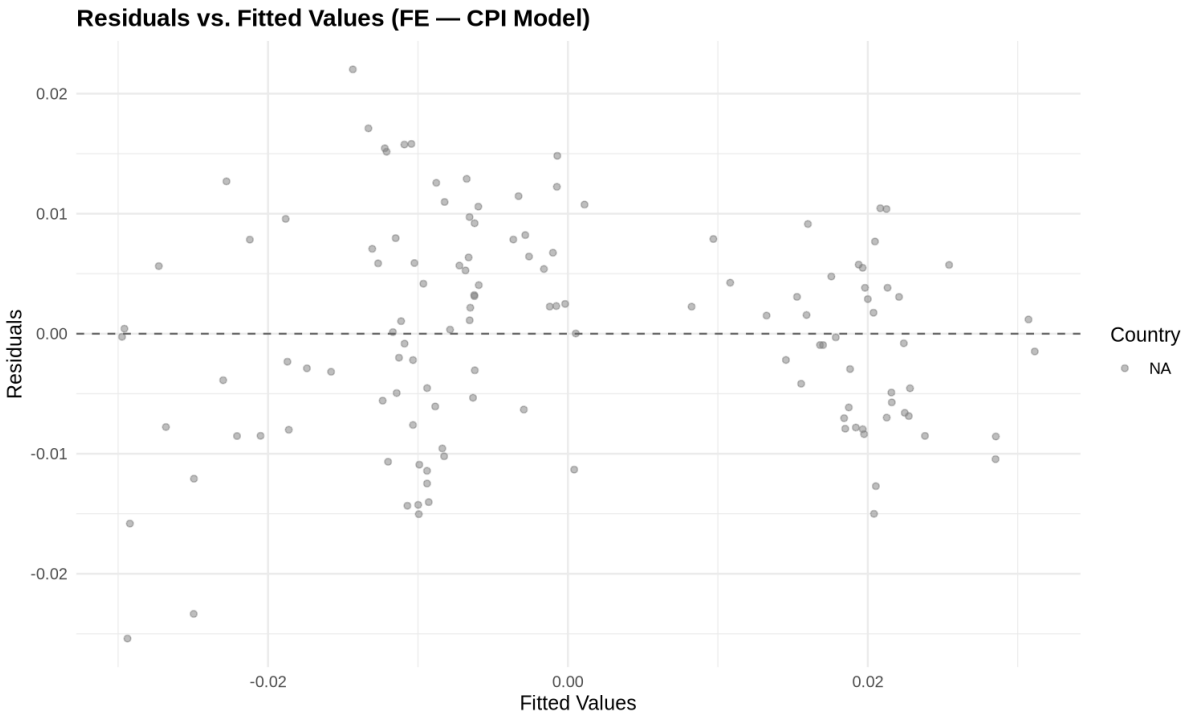


Figure A4. Residuals vs. Fitted Values (FE CPI Model)